Autonomous Planning Under Uncertainty (0970252)
Homework #3

Submission guidelines:

- Submission in pairs by: 6 December 2025, 23:59.
- Email submission as a *single zip or pdf file* to `moshiktech@gmail.com`.
- Make sure to name the submitted file ID1-ID2.zip (or .pdf) with ID1,ID2 - the students' ids.
- For hands-on parts, your source code should be submitted as well (include it in the above zip file).

Theoretical questions

1. Consider a planning session at time instant k and a given set of candidate action sequences $\mathcal{A} \doteq \{a_{k:k+L-1}\}$, where L is the planning horizon. Denote the belief over state X_{k+l} at time instant $k+l$ by

$$b_{k+l} \doteq \mathbb{P}(X_{k+l} \mid a_{k:k+l-1}, z_{1:k+l}),$$

and assume motion model $\mathbb{P}(X_t \mid X_{t-1}, a_{t-1})$ and observation model $\mathbb{P}(z \mid X)$ are given. Consider the objective function $J(b_k, a_{k:k+L-1})$ of the expected cumulative rewards, given some *belief-dependent* reward function $\rho(b, a)$ and terminal reward function $\rho_T(b)$.

   (a) Write the objective function explicitly for the above-considered setting.

   (b) Derive a recursive formulation that expresses $J(b_k, a_{k:k+L-1})$, for a given action sequence $a_{k:k+L-1} \in \mathcal{A}$, via $J(b_{k+1}, a_{k+1:k+L-1})$. Present the derivation in detail.

   (c) Consider now the corresponding value function $V_k^\pi(b_k)$ and some pre-defined policy π within the same setting. Write an explicit expression for the value function $V_k^\pi(b_k)$.

   (d) Similar to clause 1b, derive a recursive formulation that expresses $V_k^\pi(b_k)$ in terms of $V_{k+1}^\pi(b_{k+1})$.

   (e) Discuss the difference between $J(b_k, a_{k:k+L-1})$ and $V_k^\pi(b_k)$.

   (f) Consider now the *optimal* action sequence $a^\star_{k:k+L-1} = \arg\max_{a_{k:k+L-1} \in \mathcal{A}} J(b_k, a_{k:k+L-1})$. Following a similar derivation as in clause 1b, develop a recursive formulation of $\max_{a_{k:k+L-1} \in \mathcal{A}} J(b_k, a_{k:k+L-1})$ in terms of $\max_{a_{k+1:k+L-1}} J(b_{k+1}, a_{k+1:k+L-1})$. Present the derivation in detail.

Hands-on tasks

You are highly encouraged to use the Julia language for all hands-on tasks. However, this is not compulsory. In case you use Julia, the supplied skeleton may be helpful.

- **Note 1:** Your implementations might be useful in future homeworks as well, therefore it is recommended to keep them organized and well documented.
- **Note 2:** In all sections that require sampling from random distributions or random number generators, use a consistent random number generator with one of your IDs as the fixed seed from the beginning of your code, and use it in sampling methods. Make sure your code is reproducible (outputs the same results every time it runs).

In this exercise we will implement a parametric belief Bayes Filter and a simple Open Loop planning.

1. Consider the 2D environment mobile robot navigating problem from HW2, with the beacons.
Reminder: the state is only the position of the robot, i.e. $X_i \in \mathbb{R}^2$ denotes the x-y robot position at time instant i. Consider a prior over robot position at time instant 0 is available: $b(X_0) \doteq \mathbb{P}(X_0) = \mathcal{N}(\mu_0, \Sigma_0)$.

   The transition model $\mathbb{P}(X_{k+1} \mid X_k, a_k)$ is given by

   $$X_{k+1} = f(X_k, a_k) + w, \tag{1}$$

   where $w \sim \mathcal{N}(0, \Sigma_w)$ is the transition noise. We assume a linear model, i.e. $f(X_k, a_k) = F \cdot X_k + a_k$ with $F = I_{2\times 2}$ and action $a_k \in \mathbb{R}^2$ is the commanded position displacement.

   Additionally, n beacons are scatted in the environment with known locations (denoted by X_j^b for the jth beacon) and represented by a matrix $X^b \in \mathbb{R}^{2\times n}$, i.e. $X^b = [X_1^b, \dots, X_n^b]$. The robot gets a relative position observation $z^{rel} \in \mathbb{R}^2$ from a beacon if it is sufficiently close to it (distance less than d), with some measurement covariance Σ_v. For simplicity, consider the beacons are positioned sufficiently apart such that an observation from only one of the beacons can be obtained for every possible robot position. If the robot is not close enough to any beacon, it does not receive an observation (e.g., receives a null observation).

   In this exercise, we assume the accuracy of the observations z^{rel} deteriorates with the distance r of the robot from the beacon, starting from a certain distance. Specifically, for a fixed distance threshold $\mathrm{r_{min}} \leq d$, the measurement covariance for any distance $r \in [0, d]$ is:

   $$\Sigma_v(r) = (0.1 \cdot \max(r, \mathrm{r_{min}}))^2 \cdot I_{2\times 2}. \tag{2}$$

   (a) Write the corresponding observation model for the measurement z^{rel}, as a function of robot location X, beacon locations X^b, and additional parameters mentioned above.

   (b) Write the probabilistic inference for gaussian distributions, considering the models described above. Specifically, for time k, given the belief $b(X_k) = \mathcal{N}(\mu_k, \Sigma_k)$, an action a_k, and an observation z_{k+1}^{rel} (assume here it is not null), show how to calculate the propagated belief $b(X_{k+1}) = \mathcal{N}(\mu_{k+1}, \Sigma_k)$. Specify also how would this calculation change if the observation was null.

   (c) Implement a function `PropagateBelief`, which updates a Gaussian belief according to a (given) performed action, considering a recursive setting.
      - Input: a Gaussian belief from some time instant k over X_k, i.e. $b(X_k) = \mathcal{N}(\mu_k, \Sigma_k)$, and action a_k.
      - Output: an updated Gaussian belief at the next time step, i.e. $b(X_{k+1}) = \mathcal{N}(\mu_{k+1}, \Sigma_{k+1})$.

   (d) Implement a function `GenerateRangedObservationFromBeacons`, which generates an observation z^{rel} according to the observation model from clause 1a. Assume $X^b, r, \mathrm{r_{min}}$ and d are given.
      - Input: Robot location X
      - Output: Measurement z^{rel} and index of beacon within range; null otherwise.

   (e) Implement a function `PosteriorBeliefRangedBeacons`, which updates a Gaussian belief according to a (given) performed action and captured observation, considering a recursive setting.
      - Input: a Gaussian belief at instant k, i.e. $b(X_k) = \mathcal{N}(\mu_k, \Sigma_k)$, an action a_k and observation z_{k+1}^{rel}.
      - Output: an updated posterior Gaussian belief at the next time step, i.e. $b(X_{k+1}) = \mathcal{N}(\mu_{k+1}, \Sigma_{k+1})$.

(f) Consider $n = 9$ beacons equally scattered in a 9×9 grid, and let the coordinate of its bottom left corner be $(0, 0)$. Assume $b(X_0) = \mathcal{N}(\mu_0, \Sigma_0)$ with $\mu_0 = (0, 0)^T$ and $\Sigma_0 = I_{2\times 2}$, $\Sigma_w = 0.1^2 \cdot I_{2\times 2}$. Let the actual initial robot location (ground truth) be $X_0 = (-0.5, -0.2)^T$. Consider $T = 10$ and an action sequence $a_{0:T-1} = \{a_0, \ldots, a_{T-1}\}$, with $a_i \doteq (1.0, 1.0)^T$ for $i \in [0, T-1]$. Further, let $d = 1$ and $r_{min} = 0.1$.

   i. Generate the ground truth robot trajectory $\tau \doteq \{X_0, X_1, \ldots, X_T\}$ where the robot starts at X_0 and follows the given action sequence $a_{0:T-1} \doteq \{a_0, a_1, \ldots, a_{T-1}\}$.

   ii. Generate observations along the trajectory τ, by generating a single z_i^{rel} for each $X_i \in \tau$.

   iii. Calculate belief propagation along the trajectory τ while *only considering a motion model* (without observations). In other words, for each time step $i \in [0, T]$, calculate $\mathbb{P}(X_i \mid a_{0:i-1}, b(X_0)) \doteq \mathcal{N}(\mu_i^{mm}, \Sigma_i^{mm})$, considering the prior belief $b(x_0)$ and the probabilistic motion model (mm). Draw on a plot the ground truth trajectory, the beacons, and these propagated beliefs[1] (in terms of mean and covariance) at each time step.

   iv. Calculate the *posterior* beliefs along the trajectory τ, this time also *using generated observations* from clause 1(f)ii. In other words, for each time step $i \in [0, T]$, calculate $\mathbb{P}(X_i \mid a_{0:i-1}, z_{1:i}, b(X_0)) \doteq \mathcal{N}(\mu_i, \Sigma_i)$, considering the prior belief $b(X_0)$ and the probabilistic motion and observation models. Draw on a separate plot the ground truth trajectory, the beacons, and these posterior beliefs (in terms of mean and covariance) at each time step.

   v. Instead of the dynamic measurement covariance, consider a *fixed* measurement covariance $\Sigma_v = 0.1^2 \cdot I_{2\times 2}$ (measurement is still obtained only if $r \leq d$). Generate new observations along the trajectory τ for the fixed measurement covariance. Calculate the *posterior* beliefs along the trajectory τ, considering the observations from the fixed measurement covariance. Draw on a separate plot the ground truth trajectory, the beacons, and these posterior beliefs (in terms of mean and covariance) at each time step.

(g) Present in a new figure the localization estimation errors[2] $||\tilde{X}||$ versus time, for the two cases above (the dynamic 1(f)iv and the fixed 1(f)v measurement covariance).

(h) Present in a new figure the square root of the `trace()` of estimation covariance versus time for the two cases above. Compare and analyze qualitatively the obtained results.

2. Leveraging the above capabilities, let us now consider a simple *open loop planning* problem. Given a set of predefined action sequences $\mathcal{A} = \{a^j_{0:T-1}\}$ with $T = 10$, the robot has to choose the one that is expected to *minimize* the uncertainty at the final state (X_T).

   In this question we consider the information-theoretic cost function $c(b_t, a_t) = \det(\Sigma_t)$. Further, consider $M = 10$ simple action sequences in $\mathcal{A} = \{a^j_{0:T-1}\}_{j=1}^M$, where the jth action sequence $a^j_{0:T-1} = \{a^j_0, \ldots, a^j_{T-1}\}$ is defined as:

$$\forall i \in [0, T-1]: \quad a_i^j \doteq \left(1.0,\ 1.0 \cdot \frac{2 \cdot j}{M}\right)^T.$$

(a) Generate a single trajectory realization τ^j for each action sequence in $\mathcal{A} = \{a^j_{0:T-1}\}_{j=1}^N$ by following clause 1(f)i. Draw all M trajectories and beacons locations on a single plot.

(b) For each trajectory realization τ^j, generate observations along the trajectory similarly to clause 1(f)ii. Calculate the posterior beliefs along each trajectory τ^j considering the generated observations, and evaluate the terminal cost at the last time step T. Plot the terminal costs versus the trajectory index. Indicate in the plot which is the best action for the given information-theoretic cost function, and explain why this result makes sense.

[1] You can use Julia function `covellipse!` from package `StatsPlots` to draw a covariance ellipse.

[2] $\tilde{X} = \hat{X} - X$, where $\hat{X}$ is an estimate and X is the ground truth value.